Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	06 march 2026
Team ID	699d5f83fdf409fdbd1bec82
Project Name	InsureGuard — AI Insurance Fraud Detection System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	Login to the system using credentials
		Secure user session management
FR-2	Claim Submission	Customer submits insurance claim details
		Upload supporting documents
		Store claim information in database
FR-3	Data Processing	Data cleaning and preprocessing of claim data
		Feature extraction from claim dataset
FR-4	Fraud Detection	Machine learning model analyzes claim data
		Generate fraud probability score
		Identify suspicious claim patterns
FR-5	Claim Verification	Insurance officer reviews flagged claims
		Approve or reject suspicious claims
FR-6	Data Storage	Store claims and fraud results in database
		Maintain historical fraud records
FR-7	Reporting & Analytics	Generate fraud detection reports
		Provide analytics dashboard for administrators

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	System should provide a user-friendly interface for claim submission and fraud analysis.
NFR-2	Security	Sensitive user and claim data must be protected using secure authentication and encryption mechanisms.
NFR-3	Reliability	The system should provide accurate fraud detection results and operate consistently without failures.
NFR-4	Performance	Fraud detection analysis should be performed quickly to reduce claim processing time.
NFR-5	Availability	The system should be accessible to users and insurance officers whenever required.
NFR-6	Scalability	The system should handle increasing volumes of insurance claim data efficiently.